

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (CL) Examination
December 2013

CL 309 Construction Planning and Management

Date: 06.12.2013

Time: 1:30 p.m. To 4:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are the resources of construction Industries? Explain importance of each, in context with project management. 04
- (b) What is Milestone chart? How it differentiate from a Bar chart? 03
- Q - 2 (a) Construct an activity network of a project with following details. 08

Activity	1-2	2-3	2-4	3-4	3-6	4-5	5-6	5-7	6-7
Weeks	3	2	3	0	5	6	0	4	5

And

- i. Indicate Critical path.
 - ii. Find time of completion
 - iii. Calculate latest start time of each activity
- (b) What is Float? Explain different types of floats and its use in project updating. 06

OR

- Q - 2 (a) What is cost slope? Explain method of its determination with suitable example. 04
- (b) A project consists of seven activities. The time estimates in weeks of different activities are given below: 10

Activity	to	tm	tp
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

- i. Draw the project network.
- ii. Find critical activities and normal project duration.
- iii. What is probability of not completing the project within 18 weeks?

Q - 3 (a) With respect to network explain following activity times.

04

- a) Earliest start time
- b) Earliest finish time
- c) Latest start time
- d) Latest finish time

(b) Determine the minimum cost and optimum duration for the project network data given in the following Table. Indirect cost= Rs. 80 per day.

06

Activity	tn	tc	Cn	Cc
1-2	2	2	1000	1000
1-3	7	3	500	900
2-3	6	3	300	420
2-4	5	4	200	250
3-4	0	0	0	0
3-5	9	4	600	900
4-6	11	6	600	1000
5-6	6	3	700	910

(c) What are the elements of operation cost of equipment?

04

OR

Q - 3 (a) State various advantages of PERT.

03

(b) What is resource smoothing? What are the steps involved in resource smoothing?

06

(c) A construction company is planning to purchase a crane. Cost of equipment is 64 Lacs. If it is purchased under down payment of 25% at time of purchase and 10 installments of 8.5 Lacs each, calculate present worth at the rate of 18% compounded annually.

05

SECTION - II

Q - 4 (a) State factors affecting selection of construction equipment.

07

Q - 5 (a) State processes of controlling and explain project controlling techniques

04

(b) "Organization play important role for smooth and timely completion of a project". Justify the statement with relevant example.

05

(c) When the requirement/ purchase of standard equipment for a project are justified?

05

OR

Q - 5 (a) What are the circumstances under which the replacement of equipment is justified?

04

(b) What is monitoring? Why monitoring is importance of it in project management.

05

(c) An equipment is purchased for Rs. 24,000/-. If after five years its scrap value is Rs. 6000. Calculate amount of annual depreciation by any method. Rate of interest is 6%.

05

Q - 6 (a) Explain the working cycle of scarper. Which are the type of site condition, justify

06

- the use of scraper?
- (b) Write short note on compacting equipments. Discuss how soil parameters govern the selection of compacting equipment. 06
- (c) State the characteristics of Indian labours. 02

OR

- Q – 6 (a) State factors affecting the performance of equipment. 06
- (b) Define: Rimpull, Drawbar pull, Co efficient of traction, Grade resistance. 06
- (c) State the factors affecting efficiency of Indian labour. 02
